How Lean Can We Go?

ANN Pruning for Sensor Data Regression — Comparative Study of 10 Methods Across Dimensionality & Noise

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Motivation

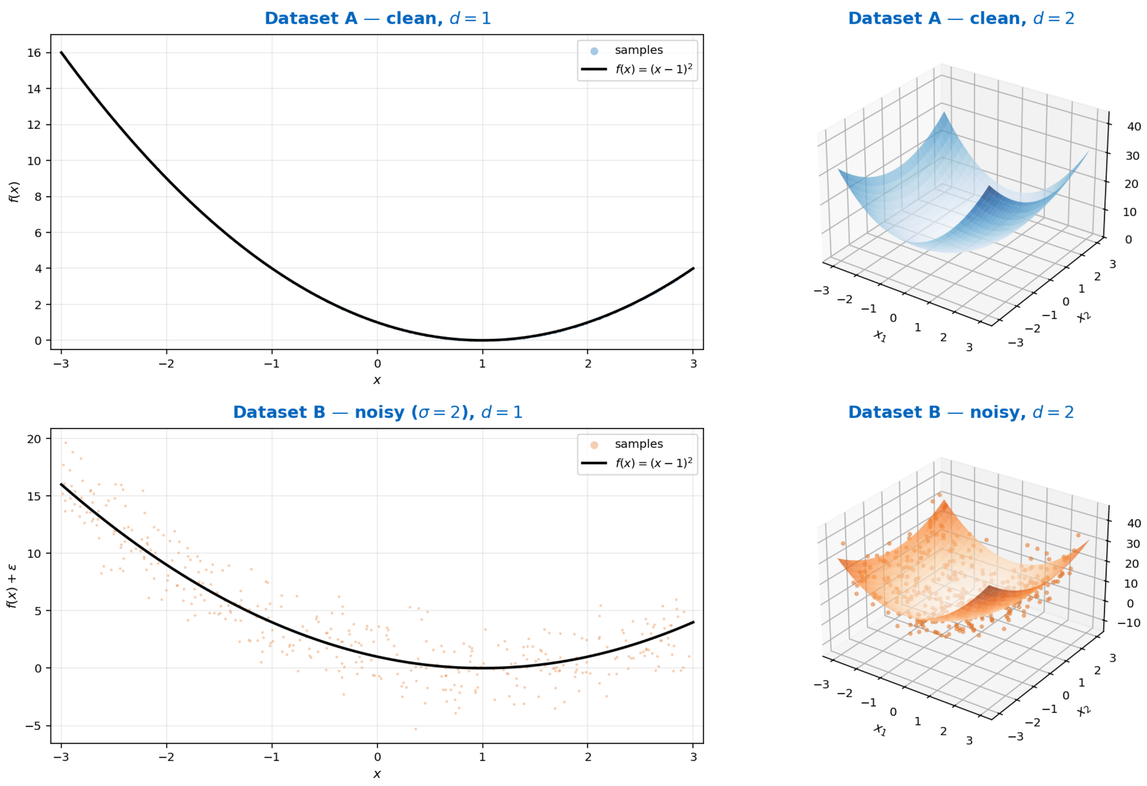
Real-world regression on thermal sensor networks demands models that are **compact**, **fast at inference**, and **interpretable** for edge deployment. Dense MLPs frequently carry substantial redundancy: many weights contribute negligibly yet consume memory and computation at every forward pass. Systematic pruning removes these weights, cutting model size and inference latency while preserving accuracy.

- **End goal:** compress MLP regressors for high-dimensional, noisy thermal sensor data.
- **Strategy:** benchmark 10 methods on controlled synthetic (quadratic) datasets where ground truth is known and noise is tunable, before scaling to sensor data.
- **Two axes of difficulty:** input dimensionality (d=1, 2, 3) and additive noise stress-test whether methods scale to harder tasks.

"How lean can we keep the network as dimensionality and noise increase?"

Datasets

Two axes of difficulty: dimensionality and noise



A (clean): $f(\mathbf{x})=(\mathbf{x}-\mathbf{1})^2$, N=5 000, $\mathbf{x}\in[-3,3]^d$. Extended to d=2, 3 via a Cholesky-based SPD quadratic. Near-zero residual is achievable; the 3-D surface panel shows the exact ground truth for d=2.

B (noisy): Same f plus $\varepsilon\sim\mathcal{N}(0,\sigma^2)$, σ=12.5% of peak-to-peak range. The noise sets an irreducible MSE floor visible in the orange scatter panels.

Methods: Pruning & Topology Discovery

Nine methods **prune** a pre-trained (or newly initialised) network by removing weights, neurons, or singular values. The tenth, **ADAGOS**, is *constructive*: it grows a compact topology from scratch, never starting from a dense network. Only OBD and OBS exploit second-order curvature information; all others are first-order or gradient-free.

Method	Removes	When?	2nd ord.	Saliency / criterion	Ref.
Post-training subtractive					
OBD	weights	post	✓	$\frac{1}{2}H_{ii}w_i^2$	[1]
OBS	weights	post	✓	$\frac{w_q^2}{2[H^{-1}]_{qq}}$, rank-1 weight correction	[2]
Magnitude	weights	post	×	$\lvert w_i\rvert$	—
Taylor FO	weights	post	×	$\lvert w_i\nabla_{w_i}\mathcal{L}\rvert$	—
Structured	neurons	post	×	$\lVert\mathbf{w}_j^{\text{out}}\rVert_2$ per neuron	—
Activation	neurons	post	×	$\text{mean}\lvert h_j(x)\rvert$ per neuron	—
Redundancy	neurons	post	×	$\lvert\rho_{ij}\rvert > 0.99$, merge correlated pairs	—
Low-Rank	sing. vals	post	×	Truncate smallest singular values of W_ℓ	—
During / before training					
SNIP	weights	at init	×	Connection sensitivity $\lvert w_i\nabla_{w_i}\mathcal{L}\rvert_{t=0}$	[5]
IMP+Rewind	weights	iterative	×	$\lvert w_i\rvert$, rewind weights to early checkpoint	[4]
Constructive — topology search (patent)					
ADAGOS	nodes/edges	joint	×	Lagrangian error estimate; topology *discovered*	[6]

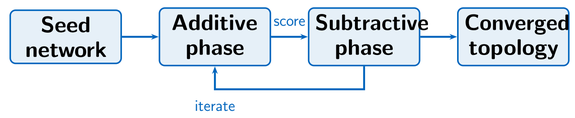
Experimental Setup

Post-training methods. Baseline MLPs selected by Optuna TPE (up to 240 trials); smallest model within 5% of best validation MSE. Fine-tuning after each pruning step follows Augasta & Kathirvalavakumar (2013) [3].

DS	d	Architecture	Params	FLOPs	Test MSE
A	1	2×16, ReLU	321	609	1.94×10^{-4}
A	2	3×32, GELU	2 241	4 385	1.35×10^{-3}
A	3	3×48, GELU	4 945	9 745	4.19×10^{-2}
B	1	1×16, ReLU	49	81	3.99
B	2	2×10, ReLU	151	281	28.4
B	3	2×4, ReLU	41	73	625

During/before-training methods. SNIP [5] masks weights at initialisation using connection sensitivity ($|w_i\nabla_{w_i}\mathcal{L}|$ at t=0) — no pre-training needed. **IMP+Rewind** [4] iterates magnitude pruning with weight rewinding to an early checkpoint (Lottery Ticket Hypothesis).

ADAGOS [6] (US12124779B2): topology discovered jointly with weights, starting from a minimal seed network:



Results: Max Sparsity at $\leq 1.5\times$ Baseline MSE

Dataset A (clean)

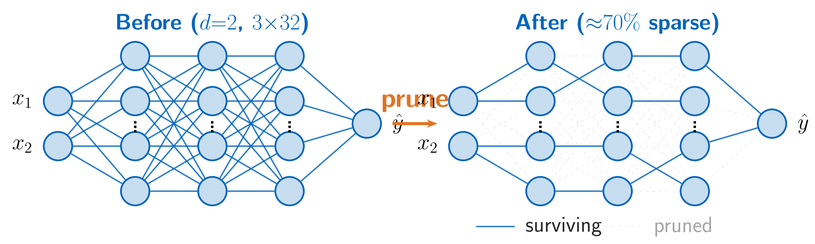
Method	d=1	d=2	d=3
OBD	**70%** 1.30× 0.32F	**30%** 1.34× 0.70F	**90%** 0.63× 0.10F
OBS	**70%** 1.01× 0.32F	—	**90%** 0.63× 0.10F
Magnitude	**60%** 1.26× 0.43F	**70%** 0.27× 0.30F	**70%** 0.37× 0.30F
Taylor FO	**50%** 1.20× 0.48F	**80%** 0.80× 0.20F	**90%** 0.61× 0.10F
SNIP	—	—	**70%** 1.47× 0.30F
IMP+Rewind	—	**70%** 0.81× 0.30F	**80%** 1.26× 0.20F
Structured	**50%** 1.43× 0.48F	**20%** 1.15× 0.80F	**40%** 1.21× 0.60F
Activation	**30%** 1.14× 0.69F	**60%** 0.92× 0.40F	**80%** 0.98× 0.20F
Redundancy	**20%** 1.27× 0.79F	**80%** 0.50× 0.21F	**70%** 0.69× 0.30F
Low-Rank	**11%** 1.01× 1.00F	**10%** 1.40× 1.00F	**10%** 1.02× 1.00F

Dataset B (noisy)

Method	d=1	d=2	d=3
OBD	**90%** 1.37× 0.31F	**80%** 1.01× 0.25F	**70%** 1.19× 0.37F
OBS	**80%** 1.06× 0.41F	**70%** 1.00× 0.32F	**50%** 1.03× 0.59F
Magnitude	**90%** 1.37× 0.28F	**80%** 1.09× 0.25F	**60%** 1.49× 0.48F
Taylor FO	**70%** 1.01× 0.43F	**90%** 1.45× 0.15F	**50%** 1.01× 0.56F
SNIP	**90%** 1.33× 0.31F	**80%** 1.43× 0.23F	**60%** 1.45× 0.42F
IMP+Rewind	**80%** 1.05× 0.38F	**90%** 1.45× 0.16F	**30%** 1.02× 0.75F
Structured	**90%** 1.24× 0.26F	**90%** 1.28× 0.15F	**40%** 1.00× 0.53F
Activation	**90%** 1.23× 0.26F	**80%** 1.43× 0.17F	**70%** 1.05× 0.32F
Redundancy	**80%** 1.37× 0.31F	**70%** 1.23× 0.34F	**50%** 1.18× 0.59F
Low-Rank	**0%** 1.00× 1.00F	**8%** 1.00× 1.00F	**12%** 1.00× 1.00F

≥70% 40–60% ≤30%/N/A **best** *Each cell: sparsity | MSE× baseline | FLOPs× baseline*

Key findings. Taylor FO and **OBD** are the top scalers on Dataset A, both reaching 90% at d=3 with MSE $\approx 0.6\times$ baseline — larger networks prove proportionally more compressible. On Dataset B the noise floor dominates: most methods achieve 80–90% sparsity at d=1, 2 with near-zero MSE penalty; **Magnitude** leads at d=1 (0.28F), **Taylor FO** at d=2 (0.15F). The **regularisation effect** on Dataset A: Magnitude at 70% sparse on d=2 achieves $0.27\times$ baseline MSE. **Low-Rank** fails in every setting; crucially, FLOPs×**1.00** even at maximum truncation confirms it provides zero inference speedup. OBS degrades at d=2 (ill-conditioned 2241×2241 Hessian).



References

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6. Bompard et al. (2024). *Method of neural network construction for the simulation of physical systems.* US12124779B2.

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